



StateEye: Intelligent Fragment-Based Web Testing Tool

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01

Problem

- Traditional web regression testing tools rely on **whole-page comparison**. They capture a full screenshot of each page and compare it to a baseline. In applications where many pages share the same navigation bar, header, and footer, this leads to **redundant checks**. Identical components are captured and validated repeatedly across pages.
- Although tools like Storybook, Chromatic support component-level testing, they depend on **manually defined components** and do not automatically infer application states or detect near-duplicate pages during crawling. Those tools are hard-coded threshold-based.

Role	Pain Point
Manual Testers	Repetitive checks, Small layout or color changes cause false failures, fatigue
Developers	Broken or outdated test suites, Minor UI changes break large portions of test suites.
QA Teams	Slow feedback, limited test coverage
Organizations	Increased cost, delayed delivery cycles

02

Proposed Solution

StateEye: An intelligent, fragment-based regression testing tool inspired by “FRAGGEN”

- StateEye addresses these limitations by introducing **fragmentation-based state abstraction** during model inference. It automatically analyzes and compares page fragments to determine state equivalence without relying on arbitrary similarity thresholds, enabling more precise models, better near-duplicate detection, and robust regression test suites suited for large-scale dynamic web applications.
- Make regression testing faster, smarter and more precise.

03

Key Features

Intelligent Web Exploration

- Simulates user actions to discover new states in web apps.
- Supports **Automated Mode**: full automatic exploration and test generation.

Fragment-Level Analysis

- Compares page sections modularly for precise detection of functional changes.
- Test oracles tolerate minor UI changes but detect real functional issues.

Near-duplicate detect

- Reduces redundant testing on near-duplicate pages

Manual Exploration

- Highlight tested vs untested areas on web pages.
- Provide guidance for next steps to testers.

Test Cases Generation

- Automatically generates stable regression test suites for future verification.

04

Technologies & Tools

Component	Technology
Browser Control	Playwright
GUI Framework	Tkinter (built-in)
Backend Language	Python 3.9+, Java, JavaScript
DOM Parsing	lxml (fastest)
Image Processing	Pillow
Tree Comparison	difflib (simple) or APTED (accurate)
Image Comparison	NumPy + Pillow
Database	SQLite
Storage	JSON + SQLite

05

Workflow

User Input

StateEye Fragment-Based Web Regression Testing ● Idle

TARGET

URL or HTML path: Browse

MODE

☒ Automated ☐ Manual

CREDENTIALS FILE Browse (optional - for login/signup)

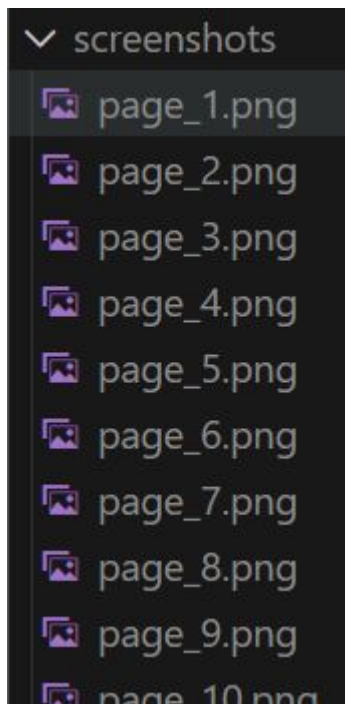
Runtime (min) **Max depth** **Max states**

▶ Start ■ Stop Generate Tests

States	Fragments	Elapsed	Classification
0	0	0s	--

[URL, Runtime, Max depth, Max states]

Crawl & Capture DOM & Screenshot

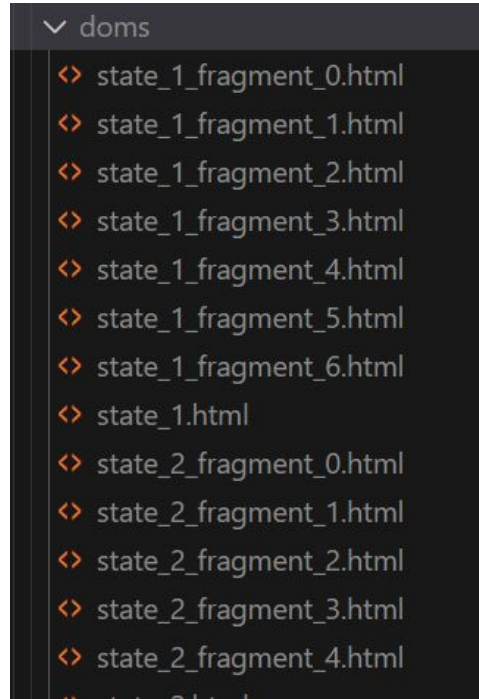
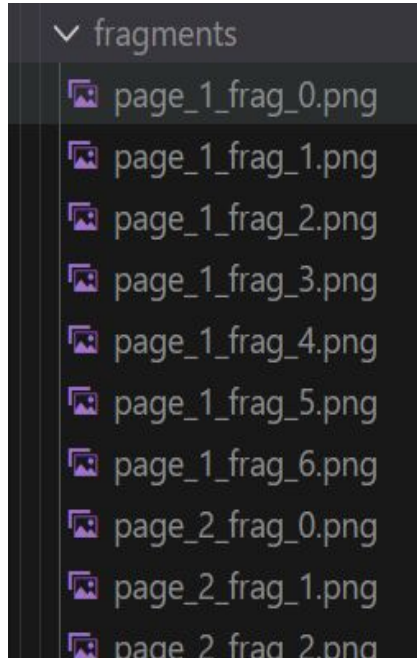


```
LIVE CRAWL OUTPUT

[00:17:53] Mode: auto
[00:17:53] Starting automated crawl...
[00:17:53] Crawl mode: hybrid
[00:17:53] Command: python -u C:\Users\sulta\Documents\StateEye\demo-...
C:/Users/sulta/Documents/StateEye/demo-...
C:\Users\sulta\Documents\StateEye\out\g...
[00:17:56] [crawl] [0m 02s] Starting cr...
[00:17:56] [crawl] [0m 02s] Max depth=5...
[00:17:57] [crawl] [0m 02s] Ready to s...
```

[Playwright]

Fragmentation (VIPS)



[Vision based page segmentation]

VIPS

- Where to Cut (HTML Tags, Block, Visual Separator)
 - N fragment identified
- Get Fragment Positions (Coordinates)
 - Fragment 1 (Header): {x: 0, y: 0, width: 1920, height: 80}
 - Fragment 2 (Logo): {x: 10, y: 10, width: 150, height: 60}
- Parent-Child Relationships
Body (root) |— Header
 |— Nav
 |— Main
- Crop images
 - Header crop: `full_image.crop((0, 0, 1920, 80))`
 - Nav crop: `full_image.crop((200, 10, 1000, 70))`
- Extract DOM (Header DOM: `<header>...</header>` (includes nav))
- Calculate hashes
 - Header DOM hash: "a3f2e1b9"
 - Header visual hash: "e7c9d2a1"

Classification

States

12

Fragments

68

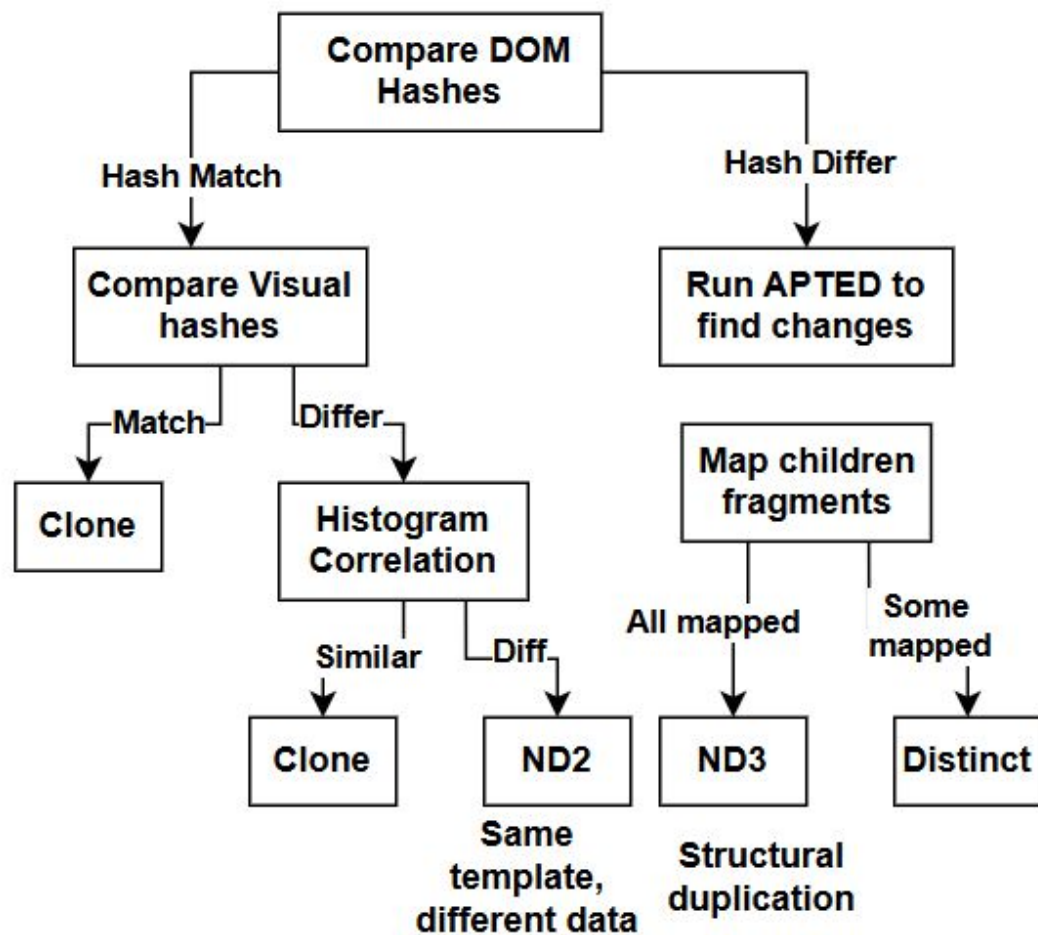
Elapsed

51s

Classification

Clone: 1 | Nd2: 4 | Nd3: 1 | Distinct: 6

```
{
  "nodes": [
    {
      "name": "state0",
      "url": "file:///C:/Users/sulta/Documents/StateEye/demo-site/index.html",
      "candidateElements": [
        {
          "top": 0,
          "left": 0,
          "xpath": "/html/body/nav[1]",
          "width": 1400,
          "height": 61
        },
        {
          "top": 93,
          "left": 244
```



Clone

Contact Us

Send a Message

Your Name

John Doe

Email Address

john@example.com

Subject

How can we help?

From Support section

Contact Us

Send a Message

Your Name

John Doe

Email Address

john@example.com

Subject

How can we help?

From Contact section



Laptop Pro 15

\$1,299

★★★★★ (127 reviews)

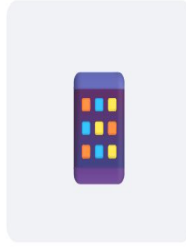
Experience the perfect blend of power and portability with the Laptop Pro 15. Featuring cutting-edge technology and a stunning display, it handles everything from everyday tasks to demanding creative workflows.

Buy Now

Back to Products

Specifications

Display	15.6 inch Retina, 2560x1600
Processor	Latest Generation Chip
Memory	16GB
Storage	512GB SSD



SmartPhone X

\$899

★★★★★ (127 reviews)

Experience the perfect blend of power and portability with the SmartPhone X. Featuring cutting-edge technology and a stunning display, it handles everything from everyday tasks to demanding creative workflows.

Buy Now

Back to Products

Specifications

Display	6.7 inch OLED, 2796x1290
Processor	Latest Generation Chip
Memory	16GB
Storage	512GB SSD

Popular Help Topics

Step-by-step guides for common issues

GUIDE

Setting Up Your Laptop

Estimated time: 15 minutes

Unbox, charge, and configure your new Laptop Pro 15. Includes Wi-Fi setup, account creation, and software updates.

Works for all Laptop Pro models purchased after 2023.

View Guide

GUIDE

Phone Data Transfer

Estimated time: 20 minutes

Transfer contacts, photos, and apps from your old phone to the new SmartPhone X using our free migration tool.

Supports transfers from both Android and iOS devices.

View Guide

TROUBLESHOOT

Tablet Won't Charge

Common fix: 5 minutes

Check the cable, clean the port, and try a hard reset. Most charging issues are resolved with these three simple steps.

If the issue persists, contact support for warranty repair.

Get Help

Current Deals

Save big on our most popular products

30% OFF

Laptop Pro 15

~~\$1,299~~

\$909

High-performance laptop with Retina display. Perfect for professionals and creatives.

Shop Now

20% OFF

SmartPhone X

~~\$899~~

\$719

AI-powered camera and 5G connectivity. The smartest phone you will ever own.

Shop Now

15% OFF

Tablet Ultra

~~\$649~~

\$552

Work and play on the go with pencil support and split-screen multitasking.

Shop Now

FREE SHIPPING

Accessories Bundle

~~\$199~~

\$149

Case, charger, and earbuds combo pack. Essential accessories for any device.

Shop Now

Distinct

Our Team



Alex Chen

CEO & Founder

15 years in tech industry. Passionate about making technology accessible to all.



Sarah

Head of Engineering

Former senior engineer at Google. Drives innovation in mobile technology.

Our Values

Quality First

Every product passes our rigorous quality testing before it reaches your hands.

Upcoming Events

MAR
15

Spring Product Launch

10:00 AM - TechShop HQ, Silicon Valley

Be the first to see our new lineup of laptops and tablets for exclusive launch-day discounts for attendees.

[Register Now](#)

APR
02

Tech Workshop: Photography with SmartPhone X

2:00 PM - Online Webinar

Join our photography experts for a hands-on workshop exploring the features of the new SmartPhone X. Learn tips and tricks for stunning mobile photography.

[Join Webinar](#)

Test case generation

- Keep Distinct & Nd3 in state graph
- Discard clone and Nd2 from state graph

```
[test] test_state_11: TechShop - SmartPhone X
      URL: file:///C:/Users/sulta/Documents/StateEye/demo-sit
      Visual distance: 2
      PASS

[test] test_state_12: TechShop - Tablet Ultra
      URL: file:///C:/Users/sulta/Documents/StateEye/demo-sit
      SKIP (clone of state #11)

=====
Results: 10 passed, 0 failed, 2 skipped (12 total)
=====
```

Manual Testing

LIVE CRAWL OUTPUT

```
[01:42:06] Mode: manual
[01:42:06] Manual mode: parsing page and launching browser.
[01:42:06] Opened http://127.0.0.1:57728/index.html
[01:42:06] Found 10 states to test
[01:42:09] [testing] a[href=products.html][text=Browse Products]
[01:42:09] [page] http://127.0.0.1:57728/products.html - 3 new states found
[01:42:10] [testing] a[href=product-laptop.html][text=View Details]
[01:42:11] [page] http://127.0.0.1:57728/product-laptop.html - 5 new states found
[01:42:12] [testing] a[href=contact.html][text=Buy Now]
[01:42:12] [page] http://127.0.0.1:57728/contact.html - 5 new states found
[01:42:14] [testing] a[href=about.html][text=About]
```


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Challenges

- VIPS Implementation
- Fragment Classification
- Threshold tuning
- Graph generation
- Map fragments

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References

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Thanks!

